

Section I**10 Marks****Attempt Questions 1–10****Allow about 20 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10

1. Let α, β and γ be roots of the equation $x^3 + 3x^2 + 4 = 0$. Which of the following polynomial equations have roots α^2, β^2 and γ^2 ?
- A. $x^3 - 9x^2 - 24x - 4 = 0$
- B. $x^3 - 9x^2 - 12x - 4 = 0$
- C. $x^3 - 9x^2 - 24x - 16 = 0$
- D. $x^3 - 9x^2 - 12x - 16 = 0$
2. If $w = \sqrt{3}\text{cis}(\pi)$, then w^4 is equal to
- A. $9\text{cis}(0)$
- B. $9\text{cis}(\pi)$
- C. $81\text{cis}(4\pi)$
- D. $4\sqrt{3}\text{cis}(\pi^4)$
3. $\int \frac{3x^2}{\sqrt{x^3 - 2}} dx$ is equal to
- A. $2\sqrt{x^3 - 2} + c$
- B. $\frac{\sqrt{x^3 - 2}}{2} + c$
- C. $\frac{2}{3(x^3 - 2)^{\frac{3}{2}}} + c$
- D. $\frac{3}{2(x^3 - 2)^{\frac{3}{2}}} + c$

4. If $y = \tan 2x$, then $\frac{d^2y}{dx^2}$ is equal to
- A. $\frac{8 \sin 2x}{\cos^3 2x}$
- B. $2 \sec^2 2x$
- C. $\frac{8 \sin 2x}{\cos 2x}$
- D. $\frac{-8 \sin 2x}{\cos 2x}$
5. A light elastic spring of natural length 2 metres hangs vertically from a ceiling. A mass of 0.5 kg is attached to the free end of the spring and hangs in equilibrium. Given that the stiffness of the spring is $0.25g$ (where g is gravity in m/s^2), then the length of the spring in metres would be
- A. $\frac{2}{g}$
- B. 2
- C. $2 + \frac{2}{g}$
- D. 4
6. The displacement x , in metres, of a particle from a fixed point at time t , in seconds, $t \geq 0$, is given by $x = 2 \cos(3t)$.
The number of oscillations made by the particle per second is
- A. $\frac{1}{\pi}$
- B. $\frac{3}{2\pi}$
- C. 2
- D. 3

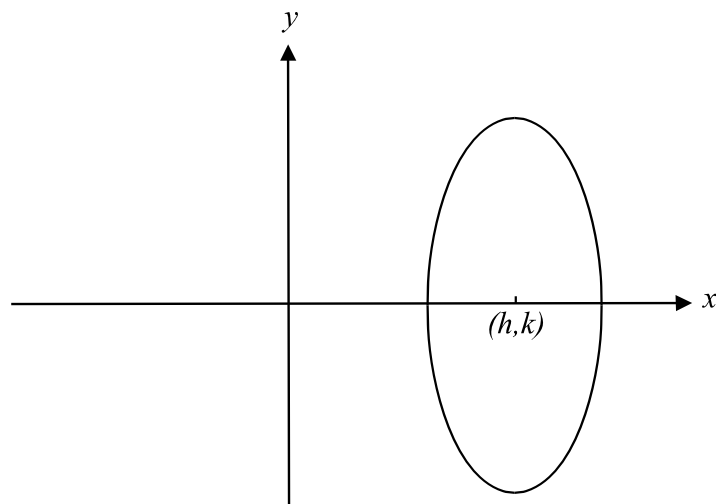
7. If $z = x + yi$, then the set of points in the complex plane given by $(2 - i)z + (2 + i)\bar{z} = 1$ is

- A. a horizontal line through $(0, 2)$
- B. a vertical line through $(-2, 0)$
- C. a straight line with gradient 2 and y-intercept -1
- D. a straight line with gradient -2 and y-intercept $\frac{1}{2}$

8. A solution to the differential equation $\frac{dy}{dx} + \frac{1}{\sqrt{1-x^2}} = 1$ is

- A. $x - \sin^{-1}x$
- B. $x - \cos^{-1}x$
- C. $x - 2\sqrt{1-x^2}$
- D. $\frac{-2x^3}{3}\sqrt{1-x^2}$

9. An ellipse is shown on the set of axes below. The scale on the x and y axes is the same. The centre of the ellipse is the point (h, k) .



The general form of the equation of an ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

For the ellipse shown above

- A. $h < k$ and $a < b$
- B. $h < k$ and $a = b$
- C. $h = k$ and $a > b$
- D. $h > k$ and $a < b$

10. The eccentricity of a hyperbola with parametric equations $x = 3 \sec \theta$ and $y = 4 \tan \theta$ is:

- A. $\frac{5}{3}$
- B. $\frac{3}{5}$
- C. $\frac{5}{4}$
- D. $\frac{4}{3}$

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 40 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available

All necessary working should be shown in every question.

Question 11 (15 marks) Use a SEPARATE writing booklet

a) Find $\int \frac{dt}{\sqrt{7+6t-t^2}}$ 2

b) i) Show that $\frac{1}{(x^2+3)(x^2+1)} = \frac{1}{2} \left[\frac{1}{(x^2+1)} - \frac{1}{(x^2+3)} \right]$ 2

ii) Hence evaluate $\int_0^1 \frac{1}{(x^2+3)(x^2+1)} dx$ 2

c) The equation $z^2 - (a+ib)z - 6i = 0$, where a and b are real, has roots α and β such that $\alpha^2 + \beta^2 = 5$.

i) Show that $a^2 - b^2 = 5$ and $ab = -6$ 2

ii) Hence find the values of a and b . 3

d) If $I_m = \int_0^k (k^2 - x^2)^m dx$, for $m \geq 1$, show that

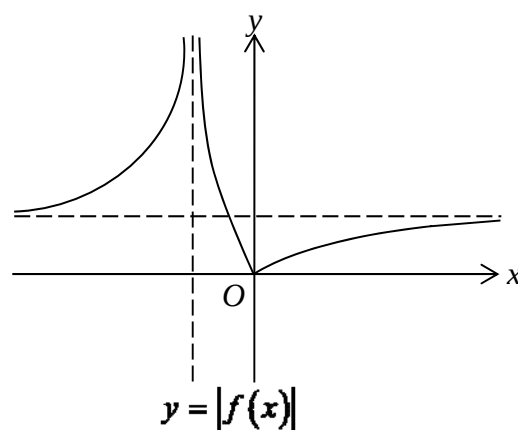
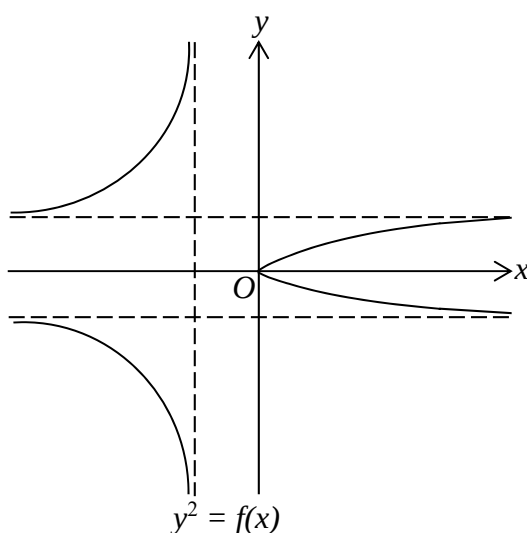
$$I_m = \frac{2k^2 m}{2m+1} \cdot I_{m-1}$$

4

$$\left[\text{Hint } \frac{x^2}{k^2 - x^2} = \frac{k^2}{k^2 - x^2} - 1 \right]$$

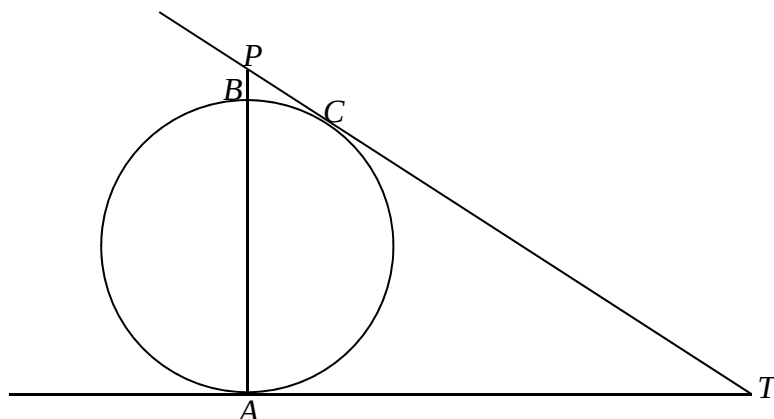
Question 12 (15 marks) Use a SEPARATE writing booklet

a) Given the graphs of $y^2 = f(x)$ and $y = |f(x)|$ as shown.



- i) Explain the shape and position of $y = f(x)$ in relation to the above graphs. 2
- ii) Hence, sketch the graph of $y = f(x)$. 2
- iii) Hence sketch the graph of $y = f(x)$. 2

- b) AB is a diameter of a circle ABC . The tangent at A and C meet at T . The lines TC and AB are produced to meet at P . Join AC and CB .



- i) Prove that $\angle BCP = 90^\circ - \angle CAT$. 2
- ii) Explain why $ATCB$ could never be a cyclic quadrilateral. 2
You must support your answers with geometrical reasons.
- c) i) Prove that if a polynomial $P(x)$ has a root α of multiplicity r , then $P'(x)$ has a root of multiplicity $(r-1)$. 2
- ii) Given that $x = 1$ is a double root of the equation $x^4 - 5x^3 + 16x^2 - 21x + 9 = 0$, and using the result of ci), or otherwise, find the other roots. 3

Question 13 (15 marks) Use a SEPARATE writing booklet

- a) i) Show that $\sin x + \sin 3x = 2 \sin 2x \cos x$. 2
- ii) Hence, or otherwise, solve $\sin x + \sin 2x + \sin 3x = 0$ for $0 \leq x \leq 2\pi$. 3
- b) A hyperbola has eccentricity $\frac{3}{2}$ and directrices $x = -4$ and $x = 4$.
Find the Cartesian equation of this hyperbola. 3

- c) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The points T and T' are the feet of the perpendiculars from the foci S and S' respectively to the tangent through P .
- i) Show that $ST = \frac{|e \cos \theta - 1|}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}}$ 2
- ii) Hence prove $ST \cdot S'T' = b^2$. 3
- d) A corner on a race track is an arc of a circle of radius 100m. The track is banked such that there is no tendency for a vehicle to move sideways when cornering at 100km/h. Find the angle, to the nearest minute, of banking. Take g to be 10 m/s^2 . 2

Question 14 (15 marks) Use a SEPARATE writing booklet

- a) If $z = \cos \theta + i \sin \theta$,
- i) Show that $z^n + z^{-n} = 2 \cos n \theta$ for $n = 1, 2, 3, \dots$ 2
- ii) Hence show that $4 \cos \theta \cos 2\theta \cos 3\theta = 1 + \cos 2\theta + \cos 4\theta + \cos 6\theta$ 3
- iii) Hence find the general solution of $\cos^2 \theta + \cos^2 2\theta + \cos^2 3\theta = 1$. 3
- b) The region bounded by the graphs of $y = x^2$ and $y = x + 2$ is revolved around the line $x = 3$. Express the volume of the resulting solid as a definite integral. Do not calculate the value of this integral. 3
- c) A solid has, as its base, the circular region in the xy plane bounded by the graph of $x^2 + y^2 = a^2$, where $a > 0$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is an equilateral triangle with one side in the base. 4

Question 15 (15 marks) Use a SEPARATE writing booklet

- a) An ellipse can be described as the locus of a point moving so that the sum of its distance from two fixed points (foci) is a constant.
- i) If the two fixed points are $A(-4,0)$ and $B(4,0)$ and the sum of the distances of $P(x,y)$ from these points is 10 units, show that the equation of the ellipse is $\frac{x^2}{25} + \frac{y^2}{9} = 1$. 2
- ii) Show that the ellipse can be represented parametrically by the equations $x = 5 \cos \theta$ and $y = 3 \sin \theta$ and find the equation, in general form of the tangent to the ellipse at $\theta = \frac{\pi}{6}$. 4
- b) The roots of $x^3 + 3px + q = 0$ are α, β and γ none of which are equal to 0.
- i) Find the monic equation with roots $\frac{\beta\gamma}{\alpha}, \frac{\alpha\gamma}{\beta}, \frac{\alpha\beta}{\gamma}$, giving the coefficients in terms of p and q . 4
- ii) Deduce if $\alpha\beta = \gamma$, then $(3p - q^2) + q = 0$. 2
- c) Given that $\sin^{-1} x, \cos^{-1} x$ and $\sin^{-1} \sqrt{1-x^2}$ are acute, show that:
- $$\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1. \quad \text{3}$$

Question 16 (15 marks) Use a SEPARATE writing booklet

- a) A particle of mass m is projected vertically upward under gravity with a speed v in a medium where resistance is mk times the speed, where k is a positive constant. If the particle reaches its greatest height H , in time T , show that $v = gT + kH$.

4

b)

Given that $\sin(x+y) + \sin(x-y) = 2\sin x \cos y$ and $\cos(x+y) + \cos(x-y) = 2\cos x \cos y$ and by using the fact that $A + B + C = \pi$ for triangle ABC .

Show that $\frac{\sin A + \sin B}{\cos A + \cos B} = \cot \frac{C}{2}$

4

- c) If $\cos \alpha = \tan x \cot y$, show that $\tan^2 \frac{1}{2} \alpha = \sin(y-x) \operatorname{cosec}(y+x)$

3

d) Evaluate $\int_0^{\frac{\pi}{3}} \frac{1}{9 - 10 \sin^2 x} dx$

4

End of Assessment Task

1 2 3 4 5 6 7 8 9 10
C A A A D B D A D A

Trial 2014 Soln Ext 2

Q1. Let $y = x^{\frac{1}{2}}$
 $y^{\frac{3}{2}} + 3y = -4$
 $\Rightarrow y^{\frac{3}{2}} = -3y - 4$
 expand + square
 $\Rightarrow y^3 - 9y^2 - 24y - 16 = 0$
 $\therefore$ (C)

A 2. ~~C~~ as $P(x)$ has real coefficients, complex roots occur in conjugate pairs.

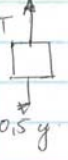
Q2 $w = \sqrt{3} \text{cis } \pi$
 $w^4 = (\sqrt{3})^4 \text{cis}(4\pi)$
 $= 9 \text{cis } 0 \therefore$ (A)

3. $\int \frac{3x^2}{12x^3 - 2} dx = \int \frac{du}{u^{\frac{1}{2}}} \cdot \frac{1}{2} du$
 let $u = 12x^3 - 2$
 $\frac{du}{dx} = 36x^2$
 $\therefore \frac{du}{dx} = 36x^2$
 $= 2u^{\frac{1}{2}} + C$
 $= 2\sqrt{12x^3 - 2} + C$ (A)

4. $y = \tan 2x$
 $\frac{dy}{dx} = 2 \sec^2 2x$
 $= 2 (\cos 2x)^{-2}$
 $\frac{d^2y}{dx^2} = -4 (\cos 2x)^{-3} \cdot 2 \sin 2x$
 $= \frac{8 \sin 2x}{\cos^3 2x}$ (A)

5.

5.


 $T = kx$
 $= 0.25gx$
 $T = 0.5g \therefore x = 2$
 Since x is the extension in the spring beyond its natural length, the length of the string is $2 + 2 = 4 \text{ m}$. (D)

(10) A
 $c = \sqrt{\frac{a^2 + b^2}{a^2}}$
 $= \sqrt{\frac{9 + 16}{9}}$
 $= \frac{5}{3}$

6. no of oscillations per sec is f
 $\therefore f = \frac{n}{2\pi}$, $n = 3$. for $x = 2 \cos 3t$
 $\therefore f = \frac{3}{2\pi}$ (B)

(7) $(2-i)z + (2+i)\bar{z} = 1$
 $2z - iz + 2\bar{z} + i\bar{z} = 1$
 If $z = x + iy$
 $\Rightarrow 2(x + iy) - i(x + iy) + 2(x - iy) + i(x - iy) = 1$
 $2x + 2iy - ix + y + 2x - 2iy + ix + y = 1$
 $4x + 2y = 1$
 $2y = -4x + 1$
 $y = -2x + \frac{1}{2}$
 find $\therefore$ (D)

(8) $\frac{dy}{dx} + \frac{1}{1-x^2} = 1$
 $\frac{dy}{dx} = 1 - \frac{1}{1-x^2}$
 $y = \int 1 - \frac{1}{1-x^2} dx$
 $= x - \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C$ (A)

(9) for centre of ellipse, $b = 0$, $h > 0$
 so $h > b$. The length of semi-major axis is b , this runs || to y axis
 semi minor is a , || to x axis.
 $\therefore a < b$ (D)

Question 11.

(a)
$$\int \frac{dt}{\sqrt{7+6t-t^2}}$$

$$= \int \frac{dt}{\sqrt{16-(t-5)^2}}$$

$$= \sin^{-1}\left(\frac{t-5}{4}\right) + C \quad (2)$$

(b) (i) Let $\frac{1}{(x^2+3)(x^2+1)} = \frac{A}{x^2+3} + \frac{B}{x^2+1}$

$\therefore 1 = A(x^2+1) + B(x^2+3)$

Let $x^2 = -1 \therefore 1 = 2B$

$B = \frac{1}{2}$

Let $x^2 = -3 \therefore 1 = A-2$

$A = \frac{1}{2}$

$\therefore \frac{1}{(x^2+3)(x^2+1)} = \frac{1}{2} \left[\frac{1}{(x^2+1)} - \frac{1}{(x^2+3)} \right]$

(ii) $\frac{1}{2} \int_0^1 \left(\frac{1}{x^2+1} - \frac{1}{x^2+3} \right) dx$

$= \frac{1}{2} \left[\tan^{-1} x - \frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \right]_0^1$

$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6} \right]$

$= \frac{\pi}{4} \left[\frac{1}{2} - \frac{\sqrt{3}}{6} \right]$

OR students can expand RHS to become LHS.

11(c) (i) $x+\beta = a+ib$

$x\beta = -ib$

$(x+\beta)^2 = x^2 + \beta^2 + 2x\beta$

$\therefore (a+ib)^2 = 5-12i$

$a^2 - b^2 + 2iab = 5-12i$

equate real + imag.

$a^2 - b^2 = 5$

$ab = -6$

(ii) $a^2 - b^2 = 5$

$a^4 - a^2b^2 = 5a^2$

$a^4 - 5a^2 - 3b^2 = 0$

$(a^2-9)(a^2-4) = 0$

$\therefore \begin{cases} a=3, b=-2 \\ \text{or } a=-3, b=2 \end{cases}$

11(d) $I_m = \int_0^k (k^2 - x^2)^m dx$

let $u = (k^2 - x^2)^m$ $v' = 1$
 $du = m(k^2 - x^2)^{m-1} (-2x) dx$ $v = x$

$\therefore I_m = \left[x(k^2 - x^2)^m \right]_0^k + \int_0^k 2mx^2(k^2 - x^2)^{m-1} dx$

$= \left[k(k^2 - k^2)^m - 0 \right] + 2m \int_0^k x^2(k^2 - x^2)^{m-1} dx$

$= 2m \int_0^k \frac{x^2(k^2 - x^2)^{m-1}}{k^2 - x^2} dx$

$= 2m k^2 \int_0^k (k^2 - x^2)^{m-1} dx - 2m \int_0^k (k^2 - x^2)^m dx$

$\therefore I_m = 2k^2 m I_{m-1} - 2m I_m$

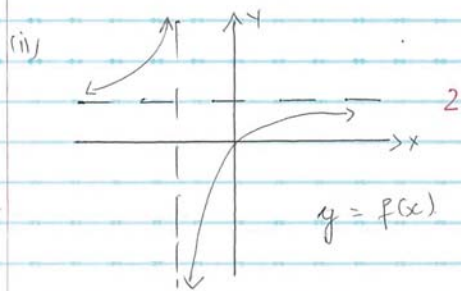
$I_m (1+2m) = 2k^2 m I_{m-1}$

$I_m = \frac{2k^2 m}{2m+1} I_{m-1}$

50% of students could not start out correctly & hence could not answer this question

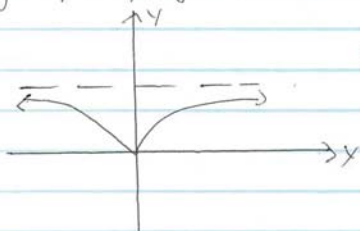
Q12

- (a) i From the graph $y^2 = f(x)$, value of $f(x)$ between vertical asymptote & y-axis is negative. (1)
 $\therefore$ the part of the graph $y = |f(x)|$ must be obtained by reflection about x-axis. (1)



- (iii) $y = |f(x)|$ is symmetrical about y-axis.
 Since $f(f(x)) = f(|x|)$
 for $x > 0$

$\therefore$ graph of $y = f(|x|)$ is



graphs answered
 ✓ well.

position wrt asymp (1)
 shape (1)

position wrt asymp (1)
 shape (1)

- 12 (b) (i) $\angle CAT = \angle ABC$ ($\angle$ bet fan + chad)
 base and $\angle BCP = \angle CAB$ (" " "
 but $\angle BAT = 90^\circ$ (diam $\perp$ radius)
 $\therefore \angle BCP = 90^\circ - \angle CAT$. (2)

✓ Reasons were
 poorly stated!

- (ii) Since $\angle BCP$ is ext $\angle$
 and its = $\angle BAC$ which is
 part of int opp $\angle$, (1) (2)
 $\triangle ABC$ cannot be cyclic quad
 as theorem ext $\angle =$ int opp $\angle$
 does not hold. (1)

✓

- (c) (i) Let $P(x) = (x-\alpha)^r Q(x)$
 $\therefore P'(x) = Q(x)[r(r-x)^{r-1}] + [(x-\alpha)^r Q'(x)]$
 ie $P'(x) = (x-\alpha)^{r-1}[rQ(x) + (x-\alpha)Q'(x)]$
 $\therefore P'(x)$ has a root
 $x = \alpha$ of multiplicity $(r-1)$ (2)
 Since $P(x) = P'(\alpha) = 0$ by subst.

Some students $P'(x)$
 but did not factorise
 $(x-\alpha)^{r-1}$ + show that
 $P'(x) = P'(\alpha) = 0$.

- (ii) Let $P(x) = x^4 - 5x^3 + 16x^2 - 21x + 9$
 $P'(x) = 4x^3 - 15x^2 + 32x - 21$

We know $P(1) = P'(1) = 0$
 $P'(1) = 4 - 15 + 32 - 21 = 0 = P'(1)$ (1)

To find other roots, several approaches
 NB $P(x) = (x-1)^2 Q(x)$

by inspect $Q(x) = x^2 - 3x + 9$
 OR Divide $(x-1)^2$ into $P(x)$ + use
 long $\div$ to find $P(x)$

OR $1, 1, \alpha, \bar{\alpha}$ are roots since (in $Q(x)$)

$\Delta < 0$

$$x^2 - 3x + 9 = 0, x = \frac{3 \pm \sqrt{27}}{2} = \frac{3 \pm 3\sqrt{3}i}{2} \quad (2)$$

✓

Q13(i). $\sin x + \sin 3x$

$$= \sin(2x-x) + \sin(2x+x)$$

$$= \sin 2x \cos x - \cos 2x \sin x$$

$$+ \sin 2x \cos x + \cos 2x \sin x$$

$$= 2 \sin 2x \cos x \quad (2)$$

(ii) $\sin x + \sin 2x + \sin 3x = 0$

$$2 \sin 2x \cos x + \sin 2x = 0$$

$$\sin 2x (2 \cos x + 1) = 0$$

$$\sin 2x = 0 \quad \cos x = -\frac{1}{2} \quad \checkmark$$

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi \quad \checkmark \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}, 3\pi, \frac{7\pi}{2}$$

b) $e = \frac{3}{2}$, directrices $x = \pm 4$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{equidistant}$$

$$x = \pm \frac{a}{e}$$

$$\therefore a = 4 \times \frac{2}{3} = \frac{8}{3} \quad \checkmark$$

$$\text{and } b^2 = a^2(e^2 - 1)$$

$$= 36 \left(\frac{9}{4} - 1 \right)$$

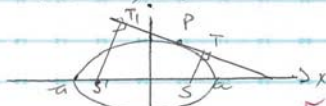
$$= 45 \quad \checkmark$$

$$\therefore \text{eqn } \frac{x^2}{36} - \frac{y^2}{45} = 1 \quad \checkmark$$

c(i) The tangent to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ has eqn } x \cos \alpha + y \sin \alpha = 1$$

$$\text{at } P(a \cos \alpha, b \sin \alpha)$$



$$S \text{ has co-ord } (ae, 0)$$

$$ST = \frac{|e \cos \alpha - 1|}{\sqrt{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}}} \quad (\perp \text{ dist from line})$$

Alt. soln

$$\sin S + \sin T = 2 \sin \left(\frac{S+T}{2} \right) \cos \left(\frac{S-T}{2} \right)$$

$$\text{LHS} = 2 \sin \left(\frac{x+3x}{2} \right) \cos \left(\frac{x-3x}{2} \right)$$

$$= 2 \sin 2x \cos (-x)$$

$$= 2 \sin 2x \cos x$$

$$= \text{RHS} \quad \text{Q.E.D.}$$

incorrect formula
 $a^2 = b^2(e^2 - 1) \quad \times$

longway
* $b \cos \alpha + a \sin \alpha - ab = 0$

$$\text{eqn tanp: } y - b \sin \alpha = -a \sin \alpha (x - ae)$$

$$-a \sin \alpha y + ab \sin^2 \alpha = -b \cos \alpha x + ab \cos^2 \alpha$$

$$ST = \frac{|b \cos \alpha (ae) + a \sin \alpha (d - ab)|}{\sqrt{b^2 \cos^2 \alpha + a^2 \sin^2 \alpha}}$$

$$= \frac{|ab e \cos \alpha - ab|}{ab \sqrt{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}}} \quad a, b > 0$$

$$ST = ab \frac{|e \cos \alpha - 1|}{ab \sqrt{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}}}$$

$$= \frac{|e \cos \alpha - 1|}{\sqrt{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}}} \quad \text{Q.E.D.}$$

13 (ii)

$$ST, S'T' = \frac{1 - e^2 \cos^2 \alpha}{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}}$$

For ellipse $b^2 = a^2(1 - e^2)$

$$\Rightarrow e^2 = 1 - \frac{b^2}{a^2} \quad \checkmark$$

$$ST, S'T' = \frac{1 - \cos^2 \alpha + \frac{b^2}{a^2} \cos^2 \alpha}{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}}$$

$$= \frac{1 - \cos^2 \alpha (1 - \frac{b^2}{a^2})}{\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2}} \quad \checkmark$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \alpha)}{b^2 \cos^2 \alpha + a^2 \sin^2 \alpha}$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \alpha)}{a^2 (1 - e^2) \cos^2 \alpha + a^2 \sin^2 \alpha}$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \alpha)}{a^2 (\cos^2 \alpha + \sin^2 \alpha) - a^2 e^2 \cos^2 \alpha}$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \alpha)}{a^2 (1 - e^2 \cos^2 \alpha)}$$

$$= b^2$$

$$= \text{RHS} \quad \text{Q.E.D.}$$



$$A = mg \sin \theta \quad B = \frac{mv^2}{r} \cos \theta$$

$$A - B = 0$$

$$\therefore \frac{mv^2}{r} \cos \theta = mg \sin \theta$$

$$\therefore \tan \theta = \frac{v^2}{gr} \quad 2.$$

$$\therefore v = \frac{100 \times 1000}{3600} \quad \checkmark$$

$$= \frac{250}{9} \text{ m/s}$$

$$\therefore \tan \theta = \frac{(250)^2}{10 \times 100}$$

$$\theta = 37^\circ 39'$$

students lost in
Alg simplification

a few could not
resolve forces
eqn $A - B = 0$.

Q14

$$\begin{aligned}
 (a) \quad (i) \quad z + z^+ &= \cos n\alpha + i \sin n\alpha \\
 &\quad + \cos(-n\alpha) + i \sin(-n\alpha) \\
 &= \cos n\alpha + i \sin n\alpha - i \sin n\alpha \quad (\text{note } \cos \text{ even} + \sin \text{ odd}) \\
 &= 2 \cos n\alpha. \quad 2.
 \end{aligned}$$

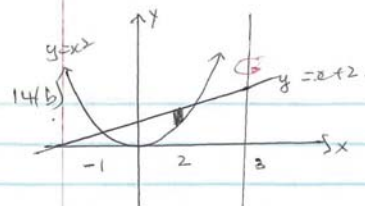
$$\begin{aligned}
 (ii) \quad z + z^{-1} &= 2 \cos \alpha, \quad z^2 + z^{-2} = 2 \cos 2\alpha \\
 z^3 + z^{-3} &= 2 \cos 3\alpha
 \end{aligned}$$

$$\begin{aligned}
 8 \cos \alpha \cos 2\alpha \cos 3\alpha &= (z + z^{-1})(z^2 + z^{-2})(z^3 + z^{-3}) \\
 &= z^6 + 1 + z^2 + z^4 + z^{-4} + z^{-2} + 1 + z^{-6} \\
 &= 2 + z^2 + z^{-2} + z^4 + z^{-4} + z^6 + z^{-6} \\
 &= 2 + 2 \cos 2\alpha + 2 \cos 4\alpha + 2 \cos 6\alpha \\
 &\quad \text{or } 4 \cos \alpha \cos 2\alpha \cos 3\alpha \quad 3. \\
 &= 1 + \cos 2\alpha + \cos 4\alpha + \cos 6\alpha
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad 2 \cos^2 \alpha + 2 \cos^2 2\alpha + 2 \cos^2 3\alpha &= 2 \\
 1 - \cos 2\alpha + \cos 4\alpha + \cos 6\alpha &= 0 \\
 4 \cos \alpha \cos 2\alpha \cos 3\alpha &= 0 \\
 \alpha &= 2k\pi \pm \pi/2 \quad \checkmark \\
 \text{or } \alpha &= \frac{2k\pi}{2} \pm \frac{\pi}{4} \quad 3. \\
 \text{or } \alpha &= \frac{2k\pi}{3} \pm \frac{\pi}{6} \quad \checkmark
 \end{aligned}$$

insufficient working.

v poorly answered

pts of intersection $x = -1, 2$.
 $\int_{-1}^2 y \cdot y \cdot dx$. Rotate strip to form a shell.

$$\therefore \text{Vol shell } dV = 2\pi r \cdot \sigma x \cdot \checkmark \quad r = 3-x$$

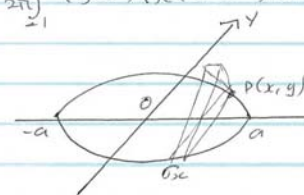
$$h = y_1 - y_2$$

$$V = \lim_{\sigma x \rightarrow 0} \sum_{x=-1}^{x=2} 2\pi (3-x)(x+2-x^2) \sigma x \quad = x+2-x^2 \quad 3$$

$$V = \lim_{\sigma x \rightarrow 0} \sum_{x=-1}^{x=2} 2\pi (3-x)(x+2-x^2) \sigma x$$

$$V = 2\pi \int_{-1}^2 (3-x)(x+2-x^2) dx$$

(c).



$$\begin{aligned}
 \Delta \text{ with sides } 2y, 2y, 2y \text{ and angle } 60^\circ \\
 dV &= \frac{1}{2} 2y \cdot 2y \cdot \sin 60^\circ \cdot \sigma x \\
 &= y^2 \sqrt{3} \sigma x \quad \checkmark
 \end{aligned}$$

$$V = \lim_{\sigma x \rightarrow 0} \sum_{x=-a}^{x=a} y^2 \sqrt{3} \sigma x$$

$$V = \lim_{\sigma x \rightarrow 0} \sum_{x=-a}^{x=a} y^2 \sqrt{3} \sigma x$$

$$= \lim_{\sigma x \rightarrow 0} \sum_{x=-a}^{x=a} (a^2 - x^2) \sqrt{3} \sigma x \quad \checkmark$$

$$= \sqrt{3} \int_{-a}^a (a^2 - x^2) dx$$

$$= 2\sqrt{3} \int_0^a (a^2 - x^2) dx$$

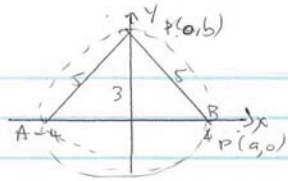
$$= 2\sqrt{3} \left[a^2 x - \frac{x^3}{3} \right]_0^a \quad \checkmark$$

$$= \frac{4\sqrt{3}}{3} a^3 \quad \checkmark$$

• answered v poorly
 • could not obtain volume ^{eqn} in terms of rotation around x=3

• answered well.

Q15
Q16



$$OP = 3 \text{ (by Pyth)} \quad \checkmark$$

$$\therefore b = 3 \quad \checkmark$$

when $P \equiv B(a,0)$, $a = 5$

$$\text{Sub } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{eqn } \frac{x^2}{25} + \frac{y^2}{9} = 1. \quad \checkmark$$

$$\begin{aligned} \text{ii) } \frac{x^2}{25} + \frac{y^2}{9} &= \frac{5^2 \cos^2 \alpha}{25} + \frac{3^2 \sin^2 \alpha}{9} \\ &= \cos^2 \alpha + \sin^2 \alpha \\ &= 1 = \text{RHS. QED.} \end{aligned} \quad \checkmark$$

$$x = 5 \cos \alpha$$

$$\frac{dx}{d\alpha} = -5 \sin \alpha$$

$$y = 3 \sin \alpha$$

$$\frac{dy}{d\alpha} = 3 \cos \alpha \quad \checkmark$$

$$\frac{dy}{dx} = \frac{dx}{d\alpha} \times \frac{d\alpha}{dy} = 3 \cos \alpha \cdot \frac{1}{5 \sin \alpha}$$

$$\text{when } \alpha = \frac{\pi}{6}$$

$$\frac{dy}{dx} = \frac{-3 \times \frac{\sqrt{3}}{2}}{5 \times \frac{1}{2}} = \frac{-3\sqrt{3}}{5}$$

$$x = 5 \times \frac{5}{2} = \frac{5\sqrt{3}}{2}$$

$$y = 3 \times \frac{1}{2} = \frac{3}{2}$$

$$\therefore y - \frac{3}{2} = \frac{-3\sqrt{3}}{5} \left(x - \frac{5\sqrt{3}}{2} \right)$$

$$3\sqrt{3}x + 5y - 30 = 0 \quad \checkmark$$

• Students used eqn of tan rather than finding it

• messy algebra.

$$\text{Q16) } x + \beta + \gamma = 0 \quad \checkmark$$

$$x\beta + x\gamma + \beta\gamma = 36 \quad \checkmark$$

$$\therefore \frac{x\gamma}{x} + \frac{x\beta}{\gamma} + \frac{\beta\gamma}{\beta} = \frac{(\beta\gamma)^2 + (x\beta)^2 + (x\gamma)^2}{x\beta\gamma} \quad \text{with}$$

$$= \frac{(\beta\gamma + x\gamma + x\beta)^2 - 2(x\beta\gamma^2 + x\beta^2\gamma + \beta^2\gamma x)}{x\beta\gamma}$$

$$= \frac{(\beta\gamma + x\gamma + x\beta)^2 - 2x\beta\gamma(\gamma + \beta + x)}{x\beta\gamma}$$

$$= \frac{(3p)^2 - 2q(0)}{-q}$$

$$= \frac{9p^2}{-q} \quad \checkmark \quad (4)$$

$$\frac{\beta\gamma}{x} + \frac{x\gamma}{\beta} + \frac{x\beta}{\gamma} = \frac{x\beta}{\gamma} + \frac{\beta\gamma}{x} + \frac{x\gamma}{\beta} = x^2 + \beta^2 + \gamma^2$$

$$\text{ii) } = (x + \beta + \gamma)^2 - 2(x\beta + x\gamma + \beta\gamma)$$

$$= 0 - 2 \cdot 3p = -6p$$

$$\therefore \frac{\beta\gamma}{x} + \frac{x\gamma}{\beta} + \frac{x\beta}{\gamma} = \frac{-6p}{x\beta\gamma} = -9$$

$$\therefore \text{reqd eqn } x^3 + \frac{9p^2}{q}x^2 - 6px + q = 0$$

$$\text{ii) } \frac{x\beta}{\gamma} = 1 \text{ is a root } \checkmark$$

$$\therefore 1 + \frac{9p^2}{q^2} - 6p + q = 0 \quad \checkmark$$

$$q + 9p^2 - 6pq + q^2 = 0$$

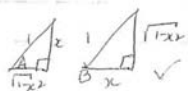
$$\therefore (3p - q)^2 + q = 0 \quad (2)$$

• very poorly answered

• Students failed to set up step 1.

• Hence could not answer (ii)

15d) Let $A = \sin^{-1} x$ & $B = \cos^{-1} x$
 $x = \sin A$ $x = \cos B$



15d)

$$\sqrt{1-x^2} = \cos A \text{ and } \sqrt{1-x^2} = \sin B$$

$$\text{LHS} = \sin(\sin^{-1} x - \cos^{-1} x)$$

$$= \sin(A-B)$$

$$= \sin A \cos B - \cos A \sin B$$

$$= x \cdot x - \sqrt{1-x^2} \sqrt{1-x^2}$$

$$= x^2 - (1-x^2) = \underline{2x^2 - 1}$$

③

Q1(a) initial position @ origin, + initial direct $\uparrow$ i.e.

$$\dot{y} = -g - kv$$

initial conditions: $t=0, x=0, v=u$

reln bet x + v

$$v \frac{dv}{dx} = -(g + kv)$$

$$-dv = \frac{v dv}{g + kv}$$

$$-k dv = \frac{k v dv}{g + kv} \quad \checkmark$$

$$-k dx = \left(1 - \frac{g}{g+kv}\right) dv$$

$$-kx + C = kv - g \ln|g+kv|$$

$$x=0, v=u \Rightarrow C = ku - g \ln|g+ku|$$

$$\therefore x = \frac{u-v}{k} + \frac{g}{k^2} \ln \left| \frac{g+ku}{g+kv} \right| \quad \checkmark \text{ (1)}$$

If particle reaches greatest height H at a time T , its speed $v=0$.

$$\therefore \text{from (1)} \\ H = \frac{u}{k} + \frac{g}{k^2} \ln \left| \frac{g}{g+ku} \right|$$

Reln bet v + t $\checkmark$

$$\frac{dv}{dt} = -(g + kv)$$

$$-dt = \frac{dv}{g+kv}, \quad -k dt = \frac{-k dv}{g+kv}$$

$$-k t = \frac{k dv}{g+kv} \quad \therefore -k t + C = \ln|g+kv|$$

$$\therefore t = -\frac{1}{k} \ln \left| \frac{g+kv}{g} \right| \quad \text{from (2)} \quad \text{(2)}$$

$$T = -\frac{1}{k} \ln \left| \frac{g}{g+ku} \right| \quad \checkmark \text{ from (2)}$$

$$\therefore \ln \left| \frac{g}{g+ku} \right| = -kT$$

$$\Rightarrow H = \frac{u}{k} - \frac{gT}{k^2}$$

$$\therefore \boxed{v = kH + gT} \quad \text{(3)}$$

answered v. poorly
only 1 stud. got
full marks.

Q1(b) Let $A = x + y$

$$B = x - y$$

$$\therefore A + B = 2x$$

$$x = \frac{(A+B)}{2}$$

$$y = \frac{(A-B)}{2}$$

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \quad \text{• students failed to apply sin to prod formula}$$

$$\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \quad \checkmark$$

$$\frac{\sin A + \sin B}{\cos A + \cos B} = \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)} \quad \checkmark$$

$$= \tan \left(\frac{A+B}{2} \right) \quad \checkmark$$

$$\text{Now } A + B + C = \pi$$

$$\therefore A + B = \pi - C$$

$$\text{LHS } \tan \frac{\pi - C}{2} = \tan \left(\frac{\pi}{2} - \frac{C}{2} \right) \quad \checkmark$$

$$= \cot \frac{C}{2}$$

$$= \text{RHS} \quad \text{Q.E.D.}$$

(4) • could not work out $\cot \frac{C}{2}$

Q1(c) Let $t = \tan \frac{x}{2}$

$$\Rightarrow \cos x = \frac{1-t^2}{1+t^2} \quad \checkmark$$

$$\therefore (1+t^2) \cos x = 1-t^2$$

$$\therefore t^2 = \frac{1-\cos x}{1+\cos x} = \frac{1-\tan x \cot y}{1+\tan x \cot y}$$

$$t^2 = \frac{1-\tan x \cot y}{1+\tan x \cot y} \times \frac{\sin y \cos x}{\sin x \cos y}$$

$$= \frac{\sin y \cos x - \sin x \cos y}{\sin y \cos x + \sin x \cos y}$$

$$= \frac{\sin(y-x) \cos(y+x)}{\sin(y+x) \cos(y-x)}$$

$$\therefore \tan^2 \frac{x}{2} = \frac{\sin(y-x) \cos(y+x)}{\sin(y+x) \cos(y-x)}$$

• Not attempted by many

$$\text{10d) } \frac{1}{9-10\sin^2 x} = \frac{\sec^2 x}{9-\tan^2 x}$$

Since $\sin^2 x = \frac{\tan^2 x}{1+\tan^2 x}$, we get

$$\frac{1}{9-10\sin^2 x} = \left[9-10 \frac{\tan^2 x}{1+\tan^2 x} \right]^{-1}$$

$$= \left[\frac{9-\tan^2 x}{1+\tan^2 x} \right]^{-1} \checkmark$$

$$= \frac{1+\tan^2 x}{9-\tan^2 x}$$

$$= \frac{\sec^2 x}{9-\tan^2 x}$$

Hence using sub $u = \tan x$
 $du = \sec^2 x dx$

$$x=0 \quad u=0 \quad \checkmark$$

$$x=\frac{\pi}{3} \quad u=\sqrt{3}$$

$$\int_0^{\pi/3} \frac{1}{9-10\sin^2 x} dx$$

$$= \int_0^{\sqrt{3}} \frac{1}{9-u^2} du \quad \checkmark$$

$$= \frac{1}{6} \left[\ln \left| \frac{3+u}{3-u} \right| \right]_0^{\sqrt{3}}$$

$$= \frac{1}{6} \ln \frac{3+\sqrt{3}}{3-\sqrt{3}} \quad 4.$$

$$= \frac{1}{6} \ln \frac{(3+\sqrt{3})^2}{3^2-(\sqrt{3})^2} = \frac{1}{6} \ln(24\sqrt{3}) \quad \checkmark$$

$$= \frac{1}{6} \ln \left(\frac{12+6\sqrt{3}}{6} \right)$$

$$= \frac{1}{6} \ln(2+3\sqrt{3})$$

or see student's
 attempt.

many few got $2/4$,
 many sub $\sin^2 x$ wrong?
 but got lost in
 Alg.
 2 ~~some~~ good attempts

Start here

Students attempt

$$\text{d. } \int_0^{\pi/3} \frac{1}{9-10\sin^2 x} dx.$$

$$= \int_0^{\pi/3} \frac{1}{9-10(1-\cos^2 x)} dx$$

$$= \int_0^{\pi/3} \frac{1}{9-10+10\cos^2 x} dx$$

$$= \int_0^{\pi/3} \frac{1}{-1+10\cos^2 x} dx.$$

$$\text{let } t = \tan x \quad \left| \begin{array}{l} x=\pi/3, t=\sqrt{3} \\ x=0, t=0 \end{array} \right.$$

$$dt = \sec^2 x dx$$

$$dt = (1+t^2) dx$$

$$dx = \frac{dt}{1+t^2}$$

$$10\cos^2 x = \frac{10}{1+t^2}$$

$$I = \int_0^{\sqrt{3}} \frac{1}{\frac{10}{1+t^2} - 1} \cdot \frac{dt}{1+t^2}$$

$$= \int_0^{\sqrt{3}} \frac{1}{\frac{10-(1+t^2)}{1+t^2}} \cdot \frac{dt}{1+t^2}$$

$$= \int_0^{\sqrt{3}} \frac{1+t^2}{9-t^2} \cdot \frac{dt}{1+t^2}$$

$$= \int_0^{\sqrt{3}} \frac{1}{9-t^2} dt \quad \rightarrow \int \frac{1}{(3-t)(3+t)} dt$$

$$= \frac{1}{t} \int_0^{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{1}{6} \left[\ln(3-t) + \ln(3+t) \right]_0^{\sqrt{3}}$$

$$= \frac{1}{6} \left[-\ln(3-\sqrt{3}) + \ln(3+\sqrt{3}) \right]$$

$$= \frac{1}{6} \ln \frac{3+\sqrt{3}}{3-\sqrt{3}}$$